SYSDESIGN REFERENCE

Design a Chat System

Requirements-to-architecture reference walkthrough.

Deliver ordered-enough, durable conversations across intermittently connected clients.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CHAT-SYSTEM

SYSDESIGN

Design a Chat System

CORE IDEA

Deliver ordered-enough, durable conversations across intermittently connected clients.

Reference architecture decision chain

01	02	03	04
Connection Management	Conversation Partitioning	Message Ids	Delivery/Read State

Decisions to defend

+ Connection management
+ conversation partitioning
+ message IDs
+ delivery/read state
+ and offline sync.

Failure modes to test

+ Duplicate delivery
+ gaps
+ fan-out spikes
+ and presence storms.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Chat System

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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